

SOLVING A PROBLEM ON HYPERBOLAS USING SIMILAR TRIANGLES

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1 Problem Statement

Problem. Consider the hyperbola given by $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Let P be any point on this hyperbola, and let the normal to the curve at P intersect the transverse axis AA' at M and the conjugate axis BB' at N . From the centre C of the hyperbola, drop a perpendicular to this normal, and denote its foot by T . Determine the value of the expression

$$\frac{PT \cdot PM}{CB^2},$$

where CB is the semi-conjugate axis of the hyperbola. (Figure 1)

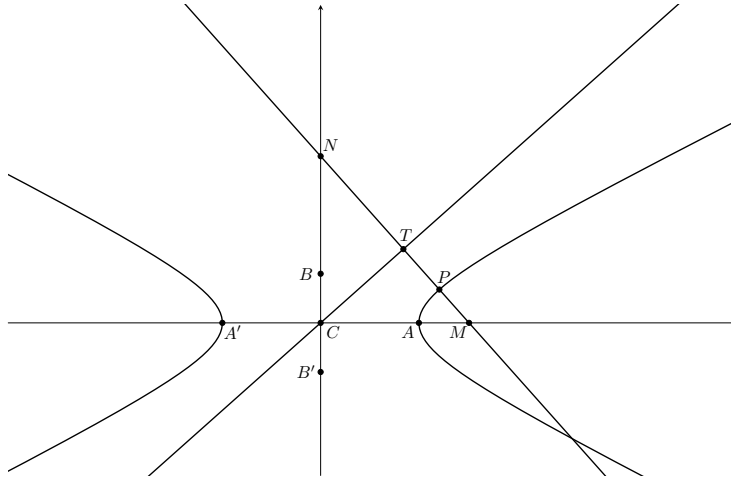


Figure 1: Determine the value of $\frac{PT \cdot PM}{CB^2}$

2 Insights and Solution Sketch

We begin by assuming that the point P lies in the first quadrant; an entirely analogous argument can be constructed if P is located in any of the remaining three quadrants.

Before proceeding to the main argument, it is helpful to examine the geometric configuration in more detail.

What is the position of the point T on the line MN ? More precisely, is T located on the segment joining P and M , or on the segment joining P and N ?

Observe that the line CT is parallel to the tangent at P to the given curve. This follows from the fact that both lines are perpendicular to the same line, the normal at P . Denote the coordinates of P by (x_1, y_1) , with $x_1, y_1 > 0$ as P is assumed to lie in the first quadrant. The equation of the tangent to the hyperbola at the point P is given by

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1.$$

Setting $y = 0$ in the above equation, we find that it meets the x -axis at

$$x = \frac{a^2}{x_1},$$

which is strictly positive since $x_1 > 0$. Hence, the tangent at P intersects the positive x -axis.

Because CT is constructed to be parallel to this tangent, the line through C parallel to the tangent must intersect the normal at some point lying between P and N on the line MN . Consequently, the point T must lie on the segment PN .

We need to find the ratio $\frac{PT \cdot PM}{CB^2}$. We know that $CB^2 = b^2$. Thus, we can try using similar triangles to determine $PT \cdot PM$.

Let $X = (x_1, 0)$ denote the foot of the perpendicular from P to the x -axis. We observe that the triangles $\triangle TCM$ and $\triangle XPM$ are similar. This similarity allows us to write the corresponding ratios of sides:

$$\frac{TC}{XP} = \frac{CM}{PM} = \frac{TM}{XM}.$$

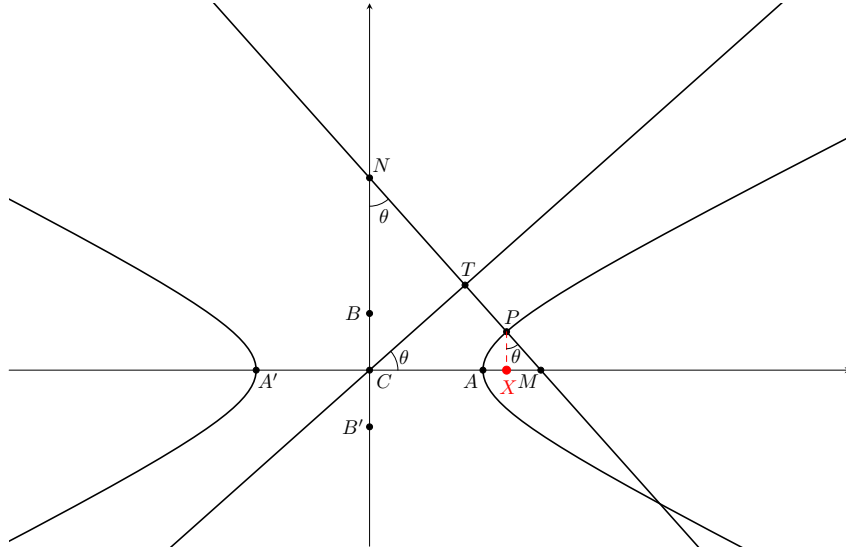


Figure 2: Point X is drawn such that $PX \perp x\text{-axis}$

3 Solution

Let X be the foot of the perpendicular from P to the x -axis, and let $P = (x_1, y_1)$. Observe that $\angle PXM = 90^\circ = \angle CTM$, and that $\angle CMT = \angle XPM$. Hence, the triangles $\triangle TCM$ and $\triangle XPM$ are similar. From this similarity, the corresponding sides are proportional:

$$\frac{CM}{PM} = \frac{TM}{XM}, \quad (1)$$

$$\implies CM \cdot XM = TM \cdot PM. \quad (2)$$

Since T lies between P and N on the line PN , we have $TM = PT + PM$. Substituting this into Eq. (2), we obtain

$$\begin{aligned} CM \cdot XM &= (PT + PM) \cdot PM \\ &= PT \cdot PM + PM^2, \\ \implies PT \cdot PM &= CM \cdot XM - PM^2. \end{aligned}$$

In the right-angled triangle $\triangle PXM$, the Pythagorean theorem gives

$$PM^2 = XP^2 + XM^2.$$

Therefore, we can rewrite the previous expression as

$$\begin{aligned} PT \cdot PM &= CM \cdot XM - XP^2 - XM^2 \\ &= XM(CM - XM) - XP^2 \\ &= XM \cdot XC - XP^2. \end{aligned}$$

Noting that $XP = y_1$ and $CX = x_1$, we have

$$PT \cdot PM = XM \cdot x_1 - y_1^2. \quad (3)$$

Next, we determine XM . The line CT is parallel to the tangent t to the hyperbola at P , whose equation is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1.$$

The slope of t is

$$m_t = \frac{b^2 x_1}{a^2 y_1},$$

and hence the slope of CT is the same. Denoting $\angle TCM = \theta$, we have

$$\tan \theta = \text{slope of } CT = \frac{b^2 x_1}{a^2 y_1}.$$

By similarity, $\angle TCM = \angle XPM = \theta$, so

$$\tan \theta = \frac{XM}{XP} = \frac{XM}{y_1}.$$

Comparing with the previous expression, we obtain

$$XM = \frac{b^2 x_1}{a^2}.$$

Substituting this into Eq. (3), we find

$$\begin{aligned} PT \cdot PM &= \frac{b^2 x_1}{a^2} \cdot x_1 - y_1^2 \\ &= \frac{b^2 x_1^2}{a^2} - y_1^2. \end{aligned}$$

Finally, dividing by $CB^2 = b^2$, the desired ratio is

$$\frac{PT \cdot PM}{CB^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}.$$

Since $P = (x_1, y_1)$ lies on the hyperbola, we have $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$. Hence, the ratio evaluates to

$$\frac{PT \cdot PM}{CB^2} = \boxed{1}.$$